

Parameter Estimation

Intuition for lower bound: poor stability of recovering uniform atomic measures from moments.

Lemma (Hundrieser et al. 2025)

There exist $s, v \in \mathcal{A}_k$ which are distinct and have $\mathbb{E}_s[x^l] = \mathbb{E}_u[x^l]$ for $1 \leq l \leq k-1$.

For $s_n = (n^{-1/2k}(\cdot))_{\#} s$, u_n same, we have: $W_1(s_n, u_n) \gtrsim n^{-\frac{1}{2k}}$

Two-point testing reduction: $\chi^2(f_{s_n}, f_{u_n}) \leq \sum_{l=1}^{\infty} \frac{(\mathbb{E}_{s_n}[x^l] - \mathbb{E}_{u_n}[x^l])^2}{l!} \lesssim \frac{1}{n}$

Parameter Estimation

Can be refined to $n^{-\frac{1}{2(k-k_0+1)}}$ when modes form k_0 well-separated clusters! (Heinrich & Kahn 2018)